H2 ANDERSON JUNIOR COLLEGE
HIGHER 2

CANDIDATE NAME

PDG

PDG INDEX NUMBER

BIOLOGY
9648/02
Paper 2 Core Paper

13 September 2016
Tuesday

2 hours

Additional Materials: Answer Paper

READ THESE INSTRUCTIONS FIRST

Write your name and PD group on all the work you hand in.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graph or rough working.
Do not use paper clips, highlighters, glue or correction fluid.

Section A
Answer **all** questions.

Section B
Answer any **one** question.

All working for numerical answers must be shown.
At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

Calculators may be used

For Examiner's Use	
Section A	**80**
1	
2	
3	
4	
5	
6	
7	
8	
Section B	**20**
8 / 9	
Total	**100**

Section A

Answer **all** the questions in this section.

1 The enzyme human immunodeficiency virus (HIV) protease is essential for the life cycle of HIV.

Fig. 1.1 shows the structure of the HIV protease. It exists as a homodimer made up of two identical polypeptide chains, each forming a subunit. The active site lies between the subunits, with each subunit contributing an aspartic acid that functions as a catalytic residue.

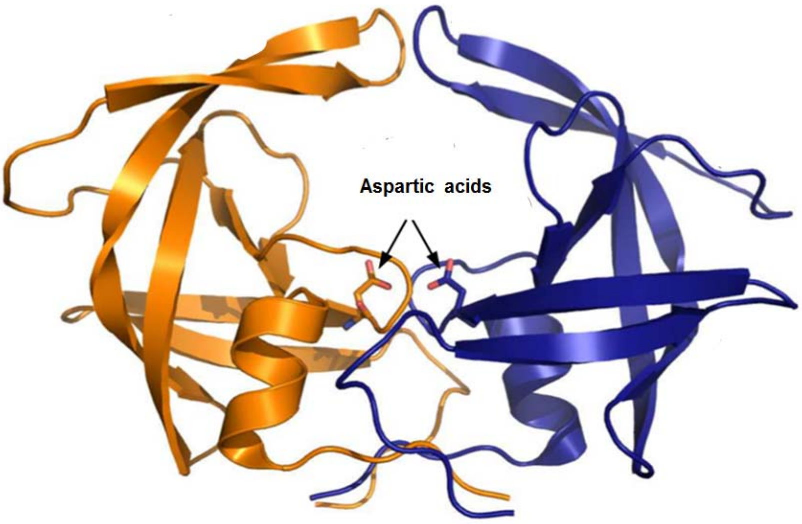

Fig. 1.1

(a) **(i)** State the function of aspartic acid in HIV protease.

- (R-group) is involved in the **hydrolysis** of **peptide bonds** (between amino acids) [1]

(ii) With reference to Fig. 1.1, outline how the structure of HIV protease is formed.

- **Polypeptide chain** consisting of **amino acids** joined by **peptide bonds**
- Within each subunit, a polypeptide chain is **folded** into (secondary structure) **alpha helices** and **beta-pleated sheets**
  Many cannot identify beta-pleated sheets shown in the diagram, hence only mentioned alpha helices.
- stabilised by **hydrogen bonds** formed between the **O atom of the CO** of one amino acid and **H atom of the NH group** of another amino acid in the **polypeptide backbone**
- Further folding of the polypeptide chain into a **globular shape / 3D shape/tertiary structure**
  structure
- maintained by **disulfide bonds, hydrogen bonds**, **ionic bonds** and **hydrophobic interactions** (name at least 2) formed between the **R groups** of the amino acids of the polypeptide chain

[4]

- The two subunit assemble together via **R-group interactions /** named bonds to form **active site**, exposing the **aspartic acids** (forming the quaternary structure)

Any 4

(iii) Suggest an advantage of protease being a homodimer.

- Shorter/smaller genome/ RNA for virus to package within its capsid
- Lesser types of tRNA / amino acids required from the host cell for synthesis of HIV protease [1]

(b) Many HIV proteases inhibitors have been developed to treat people infected with HIV. Most inhibitors act as competitive inhibitors to HIV protease. Fig. 1.2 shows three such inhibitors.

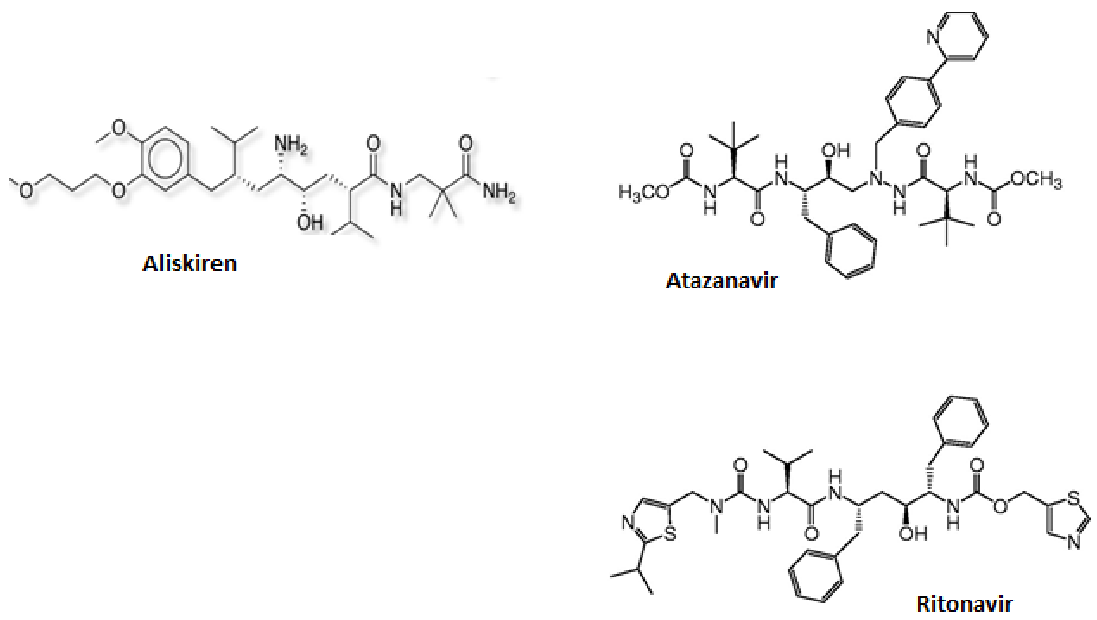

Fig. 1.2

With reference to Fig. 1.2, explain how HIV protease inhibitors work in the treatment of HIV infection.

- HIV protease inhibitors have similar structure/ shape to **polyprotein/proteins/polypeptide chain** as seen in the presence of **peptide bonds / polypeptide backbone**
- Compete with polyprotein for **binding** to **active site** of HIV protease
- Viral **<u>polyprotein</u>** not cleaved
  → functional viral(structural/ enzymatic) proteins not formed / newly formed virus **not** infective/ mature/ **functional** hence unable to infect other cells. [3]

[Total: 9]

2 Fig. 2.1 shows an electron micrograph of an actively dividing cell from *Bellevalia romana*, a flowering herbaceous plant.

A **B** **C**

Fig. 2.1

(a) **(i)** Identify stages **A** and **B**, and state the visible features that enabled your identification.

- Stage A → chromatin fibres (coil and) **condense** to become (discrete) **chromosomes** → thus (early) **prophase;**
- Stage B → chromosomes (starting to) **align singly** at **metaphase plate** → thus (early) **metaphase**; [2]

(ii) Explain the change in distance between centromeres of a chromosome in stage **C**.

- Distance between centromeres of sister chromatids **increase** in stage C;
- Because during stage C, which is anaphase, **centromere divide**;
- **Chromosomes** pulled to opposite **poles** by shortening of kinetochore microtubules/spindle fibres (with centromeres leading); [3]

(iii) Explain why it is important that replication occurs before mitosis.

- So that each **chromosome** consists of **2 genetically identical** sister chromatids (joined at centromere during prophase and metaphase);
- Each (of the two) **daughter cells** receive a copy of exact/same DNA molecule/same number and type of chromosomes → **genetically identical**; [2]

(iv) Explain how homologous chromosomes in stage **A** are genetically different from those in prophase II of the same plant.

- **Sister chromatids** of homologous chromosomes in prophase II have different alleles of the same gene as compared to those in **stage A, prophase**;

OR

- **Sister chromatids** of homologous chromosomes in prophase II are **not genetically identical** while the sister chromatids of homologous chromosomes in prophase are **genetically identical**;

- This is because in **prophase I**, (chiasmata formation and) **crossing over between non-sister chromatids of homologous chromosomes occurred**;
- Exchange of DNA segments with different alleles of the same gene/formation of new linkage [3]

groups/formation of new combinations of alleles;

Prokaryotic organisms such as *Escherichia coli* do not divide by mitosis. Apart from ribosomes, prokaryotes have no organelles comparable to those found in eukaryotes and have a circular 'chromosome' with no centromere.

(b) With reference to the information given above and your knowledge of mitosis, suggest why mitosis does not occur in prokaryotes.

- Prokaryotes do not have centrioles to form/ organise spindle fibres/ form mitotic spindles;
- Prokaryotes do not have centromeres for attachment of spindle fibres/ kinetochore microtubules (via kinetochore proteins) to separate sister chromatids; [2]

[Total: 12]

3 Fig. 3.1 is a diagram showing translation. Table 3.2 shows an mRNA codon table.

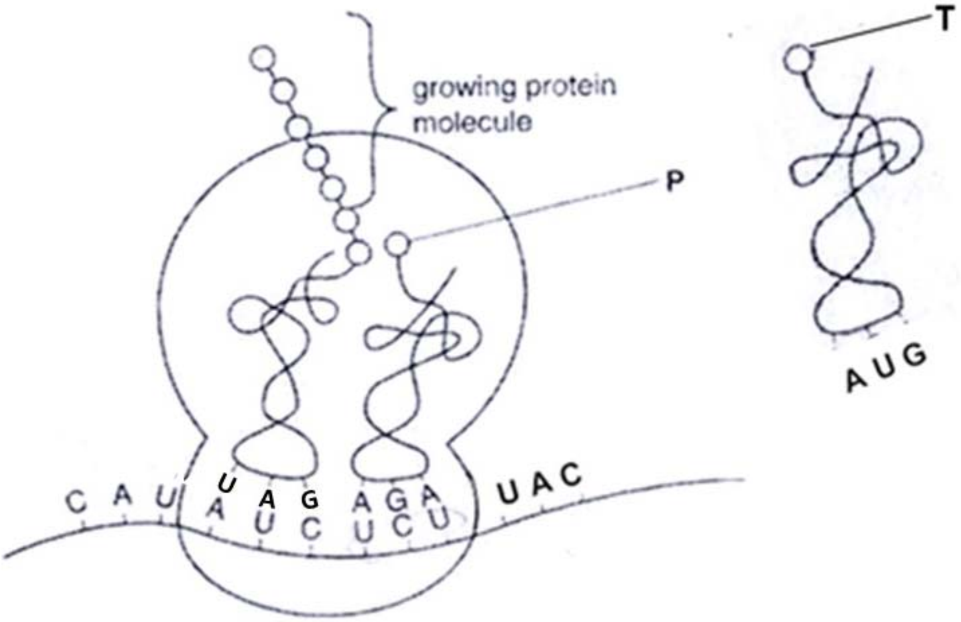

Fig. 3.1

First Base	Second Base				Third Base
	U	C	A	G	
U	phenylalanine	serine	tyrosine	cysteine	U
	phenylalanine	serine	tyrosine	cysteine	C
	leucine	serine	(stop)	(stop)	A
	leucine	serine	(stop)	tryptophan	G
C	leucine	proline	histidine	arginine	U
	leucine	proline	histidine	arginine	C
	leucine	proline	glutamine	arginine	A
	leucine	proline	glutamine	arginine	G
A	isoleucine	threonine	asparagine	serine	U
	isoleucine	threonine	asparagine	serine	C
	isoleucine	threonine	lysine	arginine	A
	(start) methionine	threonine	lysine	arginine	G
G	valine	alanine	aspartate	glycine	U
	valine	alanine	aspartate	glycine	C
	valine	alanine	glutamate	glycine	A
	valine	alanine	glutamate	glycine	G

Table 3.2

(a) **(i)** State **two** molecules required for translation that are not shown in Fig. 3.1.

- Translation **initiation** factors
- Translation **elongation** factors
- Release factors
- GTP
- ATP
- Initiator tRNA (carrying methionine)
- Aminoacyl-tRNA **synthetase**
- Peptidyltransferase

[Any 2, 0.5 marks each] [1]

(ii) Using Table 3.2 to identify **P** and **T**, briefly describe the process of translation as shown in Fig. 3.1.

- **Peptide bond formation** occurs between growing protein molecule/ last amino acid of the growing protein molecule/ isoleucine and **serine** (identify amino acid P).
- **catalysed** by **peptidyl transferase**
- ribosome translocates **1 codon/3 bases** down mRNA in **5' to 3' direction**
- tRNA carrying **tyrosine**(identify amino acid T) enters A site of ribosome
- anticodon **AUG** binds with codon **UAC**
- via (hydrogen bonds) between **complementary** base pairs/complementary base pairing.

[0.5 m each] [4]

(b) Suggest why parts of tRNA are double stranded.

rRNA is doube-stranded due to

- **Complementary base pairing** between different parts of a signle stranded tRNA.
- This contributes to the **stability** of the tRNA molecule
- allow tRNA to have a **shape** that fits into /is complementary into the E,P,A sites of ribosome/active sites of aminoacyl-tRNA sythetase. [2]

(c) State **two** ways in which DNA replication:

(i) is similar to transcription.

- Both occur in the **nucleus**;
- Both require DNA as template strand;
- Formation of **phosphodiester bonds** between **nucleotides** in DNA replication and transcription;
- **Reading** of DNA template strand is 3' to 5' direction for both DNA replication and transcription **OR Elongation** of the newly synthesis occur from 5' to 3' direction

[max 2m] [2]

(ii) differs from transcription.

- Synthesis of **a RNA strand** in transcription but synthesis of **DNA molecule** in DNA replication;
- Transcription uses **one DNA strand** as a **template** to synthesise mRNA while DNA replication uses **two DNA strands** as **template** to synthesise DNA;
- **RNA nucleotides** used as **monomers** to form polynucleotide chain in transcription while **DNA nucleotides** are used in DNA replication
- Phosphodiester bond formation/addition of monomers catalysed by DNA polymerase in DNA replication while it is catalysed by **RNA polymerase** in transcription;
- DNA replication requires a pre-existing strand known as a primer to provide a 3'OH group for polymerase to add nucleotides to while transcription can be initiated without the pre-existing strand.
- In DNA Replication, the whole DNA molecule is replicated in a single process, but in transcription only certain region/gene is being transcribed in a single process. (the phrase "in a single process" is crucial as it provides a basis of comparison)

[max 2m]

[2]

[Total: 11]

4

(a) Explain why viruses are obligate parasites.

- Related organelles/ enzymes to function: e.g lacks ribosomes for protein synthesis/ lack mitochondria for ATP generation / DNA polymerase for DNA replication

- Contains only 1 type of nucleic acid either DNA or RNA as genome but not both

any 1

AND

- hence need to takes over/ takes control/hijacks host cell metabolic machinery for the production/replication of progeny viruses [2]

Fig. 4.1 is an electron micrograph of T4 bacteriophages infecting an *Escherichia coli*.

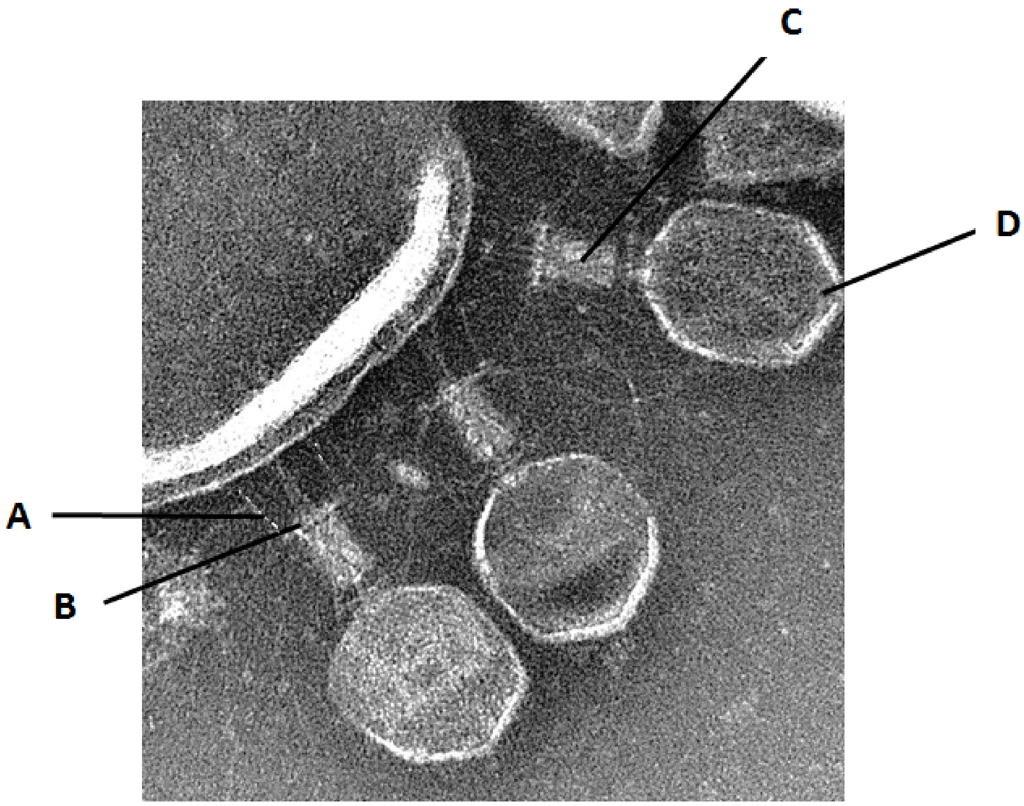

Fig. 4.1

(b) **(i)** Identify the structures labelled **A - D**.

A Tail fibre
B Base plate
C Tail/tail sheath
D Capsid head / capsid / nucleocapsid [2]

(ii) Explain why polymerase enzymes are present in human influenza virus but not in T4 bacteriophage.

- T4 phage genome is **DNA** while influenza virus genome is **RNA**
- T4 uses **host cell DNA polymerase** to replicate its genome / host cell **RNA polymerase** for transcription
  while influenza virus uses **RNA-dependent RNA polymerase** for transcription / replication which is <u>not present</u> in the host cell [2]

(c) With reference to the reproductive cycle of bacteriophages, suggest why bacteriophage infection may be beneficial to bacteria population.

- A **small piece** of a **bacteria DNA** can be incorporated into the **phage capsid** due to mistakes during viral assembly / a small region of bacterial DNA may be **excised** together with the prophage (and packaged)

- The resulting transducing phages **infect other / recipient bacteria** and newly infected cell **acquires** [3]

the original bacterial DNA

- Idea of allows **genetic recombination**/ increase **genetic variation** → increase adaptability of the bacteria to changes in environment

[Total: 9]

5 Voltage-gated sodium ion channels are integral membrane proteins that allow generation of an action potential across the axon membrane.

Fig. 5.1 shows a typical voltage-gated sodium ion channel which has two gates, the activation and inactivation gates. At different membrane potentials, the channel exists in different states depending on which gates are open or close.

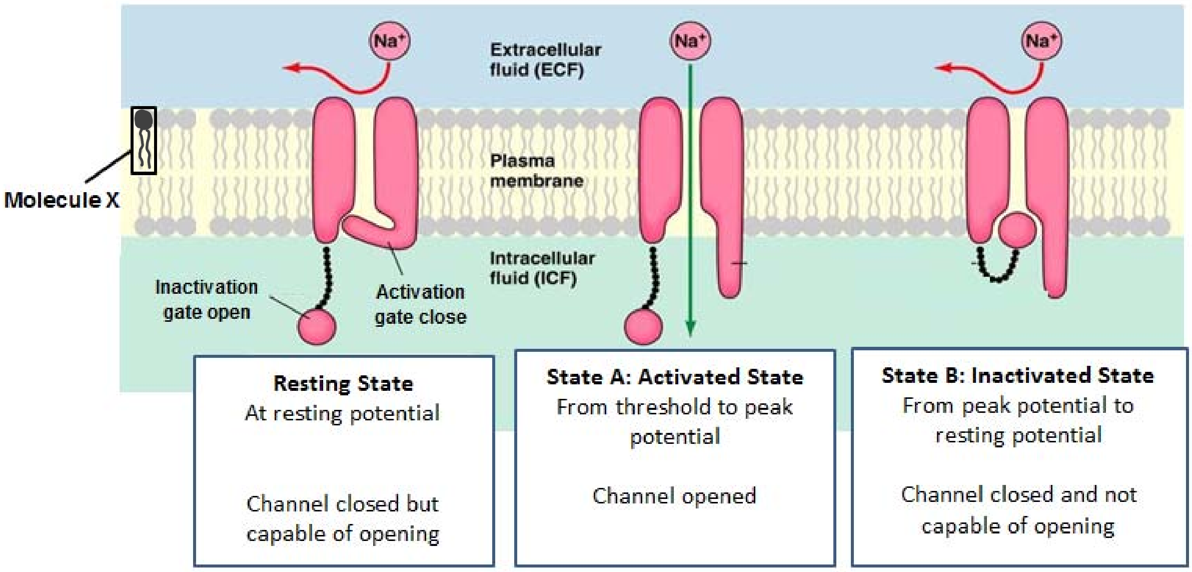

Fig. 5.1

(a) **(i)** Describe how State **A** results in depolarisation.

- In the activated state, (inactivation and) **activation gates open**
- **Na+ influx/** diffuse into the axon **through the opened ion channel**, increasing **membrane potential**

[2]

(ii) Explain how State **B** ensures unidirectional movement of nerve impulse along the neurone.

- In the inactivated state, (activation gate is open) but **inactivation gate is closed**
- this results in absolute **refractory period**
- No stimulus, can initiate another action potential / No depolarisation occurs at this part of the membrane (,resulting in the unidirectional movement of the impulse)

[3]

(iii) Describe how Molecule **X** differs from a triglyceride in terms of structure.

Molecule A (phospholipid)	Triglyceride
2 fatty acid/ hydrocarbon tails / chains	3 fatty acid/ hydrocarbon tails / chains
A phosphate group present	No phosphate group
2 ester bonds and 1 phosphoester bond	3 ester bonds

[2]

(b) Explain why triglycerides are better respiratory substrates than carbohydrates.

- release **twice as much/more** energy on oxidation/ during respiration per unit mass compared carbohydrate / compared with an **same mass** of carbohydrate

- due to two times **more hydrogen atoms** per gram/ **greater ratio of hydrogen to oxygen** atoms/ **more C-H bonds** (compared to the same mass of carbohydrate)

- Only award one mark if both answers are given but didn't mention that basis of comparison "per unit mass" at all.

[2]

[Total: 9]

6 In Drosophila, the characteristics of wing length and body colour are controlled by one gene each. Wild type flies have normal (long) wings and grey body colour while mutant flies have vestigial (undeveloped) wings and black body colour. Pure-breeding wild-type and mutant flies will breed to produce flies with normal wings and grey colour.

A male fly with wild phenotype was crossed with a female fly with mutant phenotype. The resulting offspring were as follows:

normal wings and black body colour	107
vestigial wings and grey body colour	92

Using the following symbols:

N	normal wings	**n**	vestigial wings
E	grey body colour	**e**	black body colour

(a) Draw a genetic diagram in the space below showing the cross described.

- Correct parental genotypes linked to phenotypes;
- Correct gametes;
- Correct F1 genotypes;
- Correct F1 phenotypes linked to genotypes; [4]

(b) The observed numbers are usually different from the expected numbers in any genetic cross.

(i) Suggest **two** reasons why such a difference may occur in a monohybrid cross, referring only to events after meiosis.

- sample size too small;
- chance variation / variation statistically insignificant;
- difference in survival rate of sperm / ova with particular genotypes;
- difference in survival rates of zygotes with particular genotypes; [2]

(ii) A chi-squared test was carried out on the results of the cross. A *p* value of about 0.30 is obtained.

With reference to the cross results, explain the significance of the *p* value.

- *p* value is the probability that the difference between observed and expected ratio is due to chance alone;
- since *p* is more than 0.05 / probability is more than 5%, there is no significant difference between the observed and expected ratio of 1:1;

[2]

(c) The wing length of fruit flies with normal wings varies between 70 to 85 μm.

Suggest a reason for the variation in wing length within fruit flies with normal wings.

- environmental factors such as nutrition affected the expression of allele coding for normal wings / environmental factors such as nutrition has a small effect on wing length; [1]

(d) Alleles coding for mutant phenotypes in fruit flies are caused by artificially induced mutations in laboratory bred flies. Suggest why such mutants are unlikely to be found in natural populations.

- Mutation rate is low;
- Mutant alleles may **confer selective disadvantage**, resulting in alleles being removed by **natural selection** / ref. to **decrease in frequency** of mutant alleles / unable to survive, **reproduce to pass on the alleles**;
- Mutant alleles are **recessive** and their effect on the phenotype can be **masked** by the normal dominant allele;

[max 2m] [2]

[Total:11]

7 Giant anteaters, armadillos and Australian numbats (*Myrmecobius fasciatus*) have many similar traits. This led some to believe that they were closely related and that they should be classified into the same taxon of a lower rank under the hierarchical classification system.

Table 7.1 shows the comparison of four characteristics between the three mammals

Mammal	**Characteristics**			
	Diet	**Body**	**Snout**	**Tongue**
Armadillo	Feed on insects	Covered by bony keratinised plates	Pointy snout	Long tongues
Giant Anteater	Feed on ants and termites	Covered by hair	Elongated narrow snout	Long tongues
Numbats	Feed on termites	Covered by hair	Narrow snout	Long tongues

Table 7.1

DNA sequences of selected genes such as 18s rRNA are subsequently compared between some organisms, including the three mammals, when molecular experimental techniques advanced and the results helped clarified the phylogenetic relationships of the mammals.

Fig. 7.2 shows the simplified phylogenetic tree of the organisms based on nucleotide sequence comparison results.

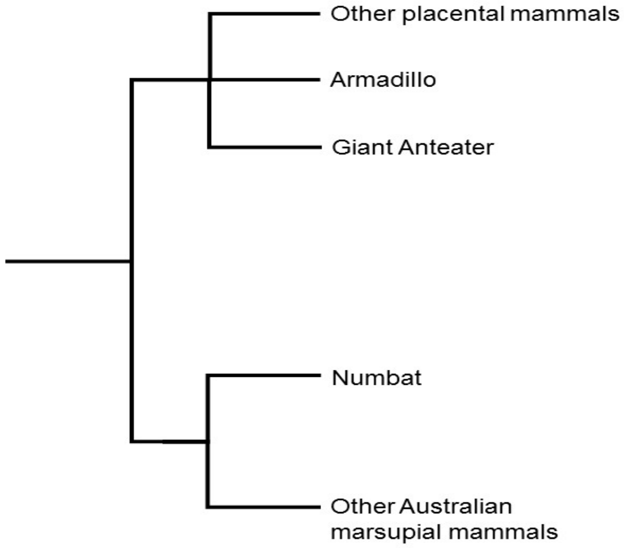

Fig. 7.2

(a) Explain the relationship between classification and phylogeny.

- Classification is organisation of species according to particular characteristics, may not take into consideration evolutionary relationship between the species;
- Phylogeny is organisation of species according to particular characteristics which takes into consideration evolutionary relationship between the species;

[2]

(b) Using the information above, explain why comparison of morphological structures led to the incorrect conclusion about the phylogenetic relationships of the three mammals.

- Structures were **analogous structures** arising from **convergent evolution**;
- All are subjected to **similar selection pressures**, e.g. same type of food (insects);
- Similar **structures** were **inherited** from **different ancestors** but selected for;
- E.g. of selective advantage: strong digging limbs to dig for insects / long tongue probe into insect nest;

[4]

(c) State **one** reason why the 18s rRNA gene was chosen to compare DNA sequences between organisms.

- 18s rRNA gene is **ubiquitous** / will be present in all organisms which serves as a good basis of comparison between organisms;
- **Essential gene** which changes very **slowly,** useful for estimating time of divergence that occurred

[1]

long time ago;

- accumulates mutations at a constant rate and therefore can be used to calibrate a molecular clock for the estimation of time of divergence between species;

[max 1m]

Fig. 7.3 shows the geographical distribution of the various mammals.

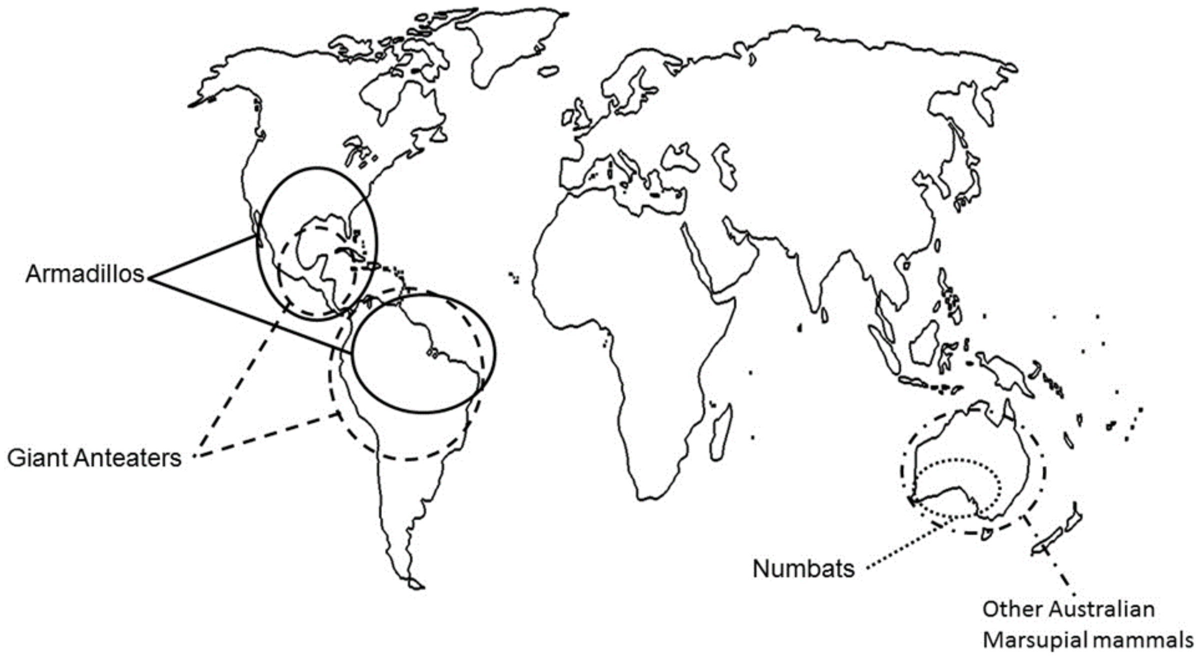

Fig. 7.3

(d) Using the information above, explain how biogeography supports the phylogenetic relationships constructed from DNA sequence comparison.

- Numbats are **more closely located** to other Australian marsupial mammals than to giant anteaters and armadillos / giant anteaters and armadillos are **more closely located**;
- Supports the phylogenetic relationship that numbats are **more closely related** to other Australian marsupial mammals than to giant anteaters and armadillos / giant anteaters and armadillos are **more closely related** / diverge from a **more recent** common ancestor;

[2]

[Total: 9]

8 Epidermal growth factor receptors (EGFR), which belong to the ErbB family of receptor tyrosine kinases, have been found to be overexpressed in many cancers, such as glioblastoma, colorectal and breast cancers.

Fig. 8.1 shows the EGFR signalling pathway.

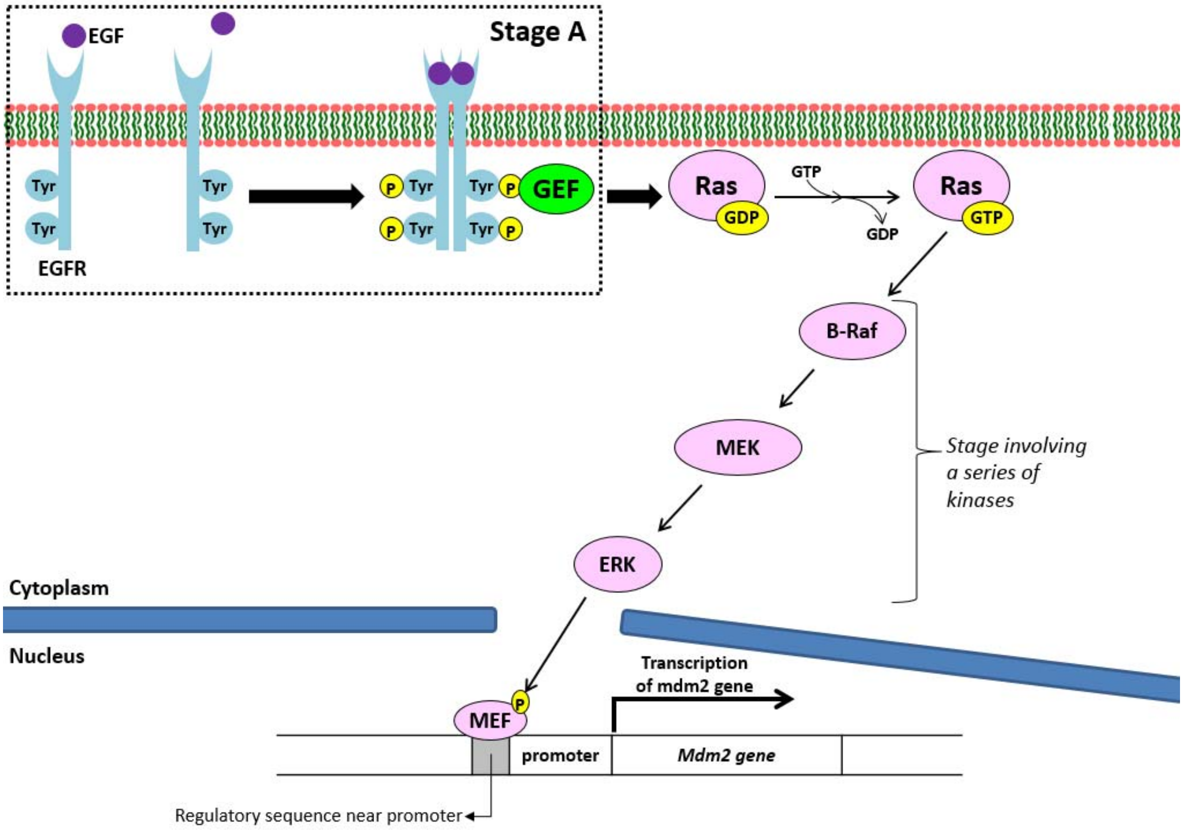

Fig. 8.1

<u>Legend:</u>

EGF	Epidermal growth factor
P	Phosphate group
GEF	Guanine nucleotide exchange factor that binds to and activates Ras
B-Raf, MEK, ERK	Kinases
MEF	A specific transcription factor that binds near the promoter

With reference to Fig. 8.1,

(a) Describe the events occurring at stage A.

- Epidermal growth factor (EGF) binds to **extracellular ligand-binding site** of epidermal growth factor receptor (EGFR) via complementary shape;
- leads to **dimerisation** of EGFR subunits occur/EGFR subunits aggregate to **form dimer**, causing the tyrosine kinase region on each EGFR subunit to be (exposed and) **activated**;
- **Autophosphorylation** occurs, each activated tyrosine kinase region **cross-phosphorylates** the other receptor at the tyrosine residues on their cytoplasmic/intracellular domain;
- (GEF binds to phosphorylated tyrosine kinase residues of fully-activated EGFR and) GEF is **activated** by phosphorylation; [4]

(b) **(i)** Suggest how a mutation in Ras GTPase that causes GTP to be permanently bound results in the overexpression of mdm2.

- Mutation of (intrinsic) GTPase of Ras → GTP bound to Ras cannot be hydrolysed to GDP (→ Ras remains/always active/hyperactive);
- Kinase B-Raf is **always activated** by Ras, and kinases MEK and ERK are always phosphorylated and active/resulting in **continuous** trigger of **phosphorylation cascade**;
- (Specific) transcription factor MEF is always phosphorylated and thus is able to bind to the proximal control element (via complementary shape)
- accelerating and stabilising formation of transcription initiation complex and hence a high rate of transcription of mdm2;

[Any 3] [3]

(ii) Mdm2 is an enzyme which catalyses the addition of ubiquitin to p53. Explain how high levels of mdm2 enzyme may lead to increased chances of cancerous growth.

- Ubiquitinated p53 degraded/hydrolysed/broken down (by proteasome into short peptides);
- Absence of p53 (transcription factor binding to DNA/control elements) to **trigger transcription of genes** that code for proteins involved in cell cycle arrest/DNA repair/apoptosis;
- (Fail to inhibit the cell cycle /no DNA repair mechanism / allow evasion of mutated cells from apoptosis□→ accumulation of mutations in other proto-oncogene/ tumour suppressor genes to occur) → uncontrolled cell division. [3]

[Total: 10]

Section B

Answer **EITHER 9 or 10.**

Write your answers on the separate answer paper provided.
Your answer should be illustrated by large, clearly labeled diagrams, where appropriate.
Your answers must be in continuous prose, where appropriate.
Your answers must be set out in section (a), (b) etc., as indicated in the question.

9 **(a)** Explain how fluidity of biological membranes can be maintained and the importance of fluidity to membrane function. [8]

How fluidity is maintained (max 4)

- Definition of fluidity: phospholipids and proteins free to move laterally within membrane/ transversely (phospholipids only)

- Unsaturated hydrocarbon/ fatty acid tails / C=C in hydrocarbon tails have **kinks** which keep the phospholipid molecules in the membrane from **packing close together** (enhancing membrane fluidity) (Accept reverse argument)

- **Cholesterol** found in between phospholipids/ embedded in membrane prevents / disrupts close/ regular packing of phospholipids in the membrane at low temperature

- **Hydrophobic interactions** between cholesterol and phospholipid tails restrict phospholipid movement at high temperature

- Hence cholesterol preventing membrane from freezing at lower temperatures and membrane prevented from being overly fluid / integrity of the membrane is maintained at high temperature

- Membranes freeze at a lower temperature/ has more fluidity if it has a higher proportion of phospholipids with unsaturated hydrocarbon tails and cholesterol (accept converse) reject unsaturated phospholipids

- Accept longer length of saturated fatty acid chains ensures more hydrophobic interactions between them → membrane less fluid

Importance of fluidity to membrane function (max 4) :

- ref. to need for **invagination / pinching in** of the cell surface membrane during cytokinesis

- ref. to need for **invagination / pinching in** of the cell surface membrane during endocytosis/ pinocytosis

- ref to need forformation of **pseudopodia** in cell surface membrane to engulf substances phagocytosis

- ref. to **budding/ pinching off vesicles** for the
    - transport / trafficking of proteins from rough ER to Golgi apparatus
    - from *trans* face of the Golgi apparatus during protein sorting to outside of the cell / / other membrane-bound organelles

- ref. to **fusion of vesicles membrane**
  - from endoplasmic reticulum / containing proteins to (cis-face)Golgi apparatus for protein modification;
  - with cell surface membrane to release secretory molecules from the cell / exocytosis;
  - fusion of endocytotic vesicles with lysosomes for digestion

- ref. to membrane fluidity allowing **movement of protein molecules embedded** on the cell surface membrane for dimerisation of receptors;

- ref. diffusion of small or non-polar molecules through the **gaps/transient pores** (between the phospholipids) in membrane, e.g. water / oxygen / carbon dioxide

- ref. to fluidity important for transmembrane transport **proteins changing conformation,** such as sodium-potassium pump / ligand-gated sodium ion channel / voltage-gated sodium ion channel (at least one named example);

- accept fluidity required for **embedment of protein molecules** such as ETC in thylakoid membrane / channel proteins on cell surface membrane for transport.

(b) Plant cells have a cellulose cell wall outside the cell surface membrane. Explain how the structure of cellulose is related to its function. [7]

- Each cellulose **chain/molecule** is made up of **large number** of **β-glucose** monomers
- giving a **long/ large** molecule which is **insoluble** in water
- **Alternate/ successive/neighboring** glucose monomers are being **inverted/rotated 180°** relative to one another to allow the formation of β**(1, 4) glycosidic bonds**,
- resulting in a **straight / linear** cellulose chain.
- **Straight parallel chains with OH-groups projecting in all directions** from each chain, allowing large number of **hydrogen bonds** to **cross -links** neighboring chains.
- allowing **bundling of cellulose chains** into microfibrils, macrofibrils and fibres
- resulting in **high tensile strength**
- in plant cell wall, this prevents plant cells from bursting when placed in solutions of high water potential.
- Maintain the **shape** of the cell/protect cells from physical and mechanical injury
- Cellulose has **large intermolecular spaces between macrofibrils**
- allows the passage of water and solute molecules through the plant cell walls (fully permeable)

(c) Describe how photophosphorylation differs from oxidative phosphorylation. [5]

Features	Oxidative phosphorylation	Photophosphorylation
Location	inner mitochondrial membrane	thylakoid membrane of chloroplast
Functions in the presence of	oxygen	light
Source of energy	(from oxidation of) glucose	light

No. of electron transport chain	one	two
Electron flow	linear - one-way	linear or cyclic
Final electron acceptor	oxygen	NADP (non-cyclic) P700 (cyclic)
Involvement of water	water produced	photolysis of water
Establishment of proton gradient	protons pumped outwards from matrix across inner mitochondrial membrane into intermembrane space	protons pumped inwards from stroma across thylakoid membrane into thylakoid space
Products	ATP, water	ATP (cyclic) ATP, NADPH and oxygen (non-cyclic)

[Total: 20]

Or

10 **(a)** Distinguish between gene mutation and chromosome structural mutation. [4]

Gene mutation	Chromosomal mutation
Change in structure of DNA or nucleotide / DNA sequence of a gene / a single locus on a chromosome	Change in chromosome structure / DNA / nucleotide sequence of gene (mostly) unchanged
Caused by deletion, insertion, substitution or inversion of one / several bases / nucleotides	Deletion, inversion, translocation or duplication of chromosomal fragments / several gene loci
Give rise to new alleles	Rearrangement of loci of genes / alleles / reshuffling / recombination / new combination of alleles
More frequent than chromosomal mutations because genes outnumber chromosomes by several thousand to one	Less frequent
Play more important role in evolution than chromosomal mutations because new alleles increases gene pool for natural selection to operate	Play a less important role in evolution than gene mutations because chromosomal mutations involve only reshuffling of alleles that already exist in gene pool.

(b) Describe how the most common CFTR gene mutation affects function of the protein and explain why other mutations vary in the extent to which they affect protein function. [8]

How the most common CFTR gene mutation affects function of protein

- **Deletion** of 3 bases **GAA** from CFTR gene on **chromosome 7**

- Net effect: results in **loss** of amino acid **phenylalanine** (position 508) in the polypeptide chain;

- Altering / affecting **R-group interactions** of protein→ **tertiary structure** of CFTR changed

- Mutant CFTR cannot allow chloride ions to diffuse out of the cells / degraded therefore cannot (serve as channel protein to) allow diffusion of chloride ions out of the cell;

Why other mutations vary in the extent to which they affect protein function (max 5)

Non-function proteins

- **Base addition / deletion** (not in multiple of threes) results in **frameshift mutation**
- All nucleotides downstream of mutation will be improperly wronged into codons
  Accept frameshift mutation leading to nonsense mutation
- Drastic change in tertiary structure→ non-functional proteins

- **Base substitution/ addition / deletion** resulting in **nonsense mutation**
- resulting in codon coding for amino acid changing to **stop codon** → translation terminated
- resulting in **truncated protein** → non-functional proteins

Little effect / some change in function

- **Base substitution/ addition / deletion** results in **missense mutation**

Either

- Codon encode for a **different amino acid** of **different R-group property** as original amino acid or amino acid change occur at essential region
- **change** in **tertiary structure** of CFTR protein→ less functional CFTR

OR

- Codon encode for a **different amino acid** of **similar R-group property** as original amino acid or Amino acid changed at a **non-essential region** of the protein

- **Little/small change** in **tertiary structure** of CFTR protein → little change/ small effect in function

No change in function

- **Base substitution** on the **wobble base** of a codon results in **silent** mutation
- will lead to altered codon coding for the **same amino acid**
- **Tertiary structure** of CFTR protein not changed, function not changed

OR

- Mutation that occurs in CFTR gene occurs in **non-coding introns**
- Does not change the DNA sequence that is eventually expressed / mRNA sequence that is eventually translated
- No change in tertiary structure → function not changed

(c) With reference to mutation of named genes, outline the development of cancer. [8]

Max 6 for outlining the development of cancer.

- Cancer involves **uncontrolled cell division;**
- Mutation first occurs to **single cell**, leading to **abnormal proliferation of single cell** into (clonally derived / **genetically identical tumour cells;)**
- Normal cells are converted to cancer cells by **accumulation of several mutations;**
- **Daughter cells** of a cell bearing such mutations will **become dominant within tumour cell population;**
- **Gain of function mutations** of **proto-oncogenes**, converted to **oncogenes**;
- Proteins encoded in oncogenes **produced in excessive amounts** / **high expression of oncogenes** producing high levels of proteins or **Hyperactive proteins** may be produced (which are constitutively active or resistant to degradation)
- **Loss of function mutations** of **tumour suppressor genes**;
- **No functional/inactive tumour suppressor proteins**
- **Deactivation** of genes involved in **cell-to-cell adhesion**, resulting in loss of anchorage dependence / tissue invasion and **metastasis**;
- **Activation** of genes involved in **angiogenesis,** resulting in formation of blood vessels that provide cancer cells with nutrients and oxygen;

Max 2 for named genes.

- E.g. of proto-oncogene: mutataion of Ras gene, involved in **signalling pathway / signal transduction** that results in **stimulation of cell cycle → cell division.**
- E.g. of proto-oncogene: mutation of c-myc, **transcription factor** that regulates expression of genes involved in cell division;
- E.g. of tumour suppressor gene: mutation of p53 gene, which encodes for a **transcription factor** which stimulation transcription of genes involved in **cell cycle arrest**, **DNA repair** and **initiation of apoptosis** when there is **DNA damage**.
- Activation of **telomerase gene**, allowing telomerase to **maintain length of telomeres →** Preventing the cell from entering **replicative senescence**/ **divide unlimited number of times**;

[Total:20]